

5) Explain the Friedman's quantity theory of money.

Demand for money

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5.5 LET US SUM UP

Money demand refers to the desire of individuals and institutions to hold cash and other liquid assets for various purposes. It plays a crucial role in determining the overall demand for money within an economy, which in turn influences monetary policy decisions and economic stability.

Factors Affecting Money Demand

Interest rate : Interest rate is the price of demanding money. Holding money carries an opportunity cost in the form of interest lost which could be earned if the money demanded were invested in some other assets like bond. Thus, there is a negative relationship between the demand for money and interest rate.

Income (or GDP) : Higher the level of income means higher the volume of transactions, therefore, there is a positive relationship between the income and demand form money.

Price Level: The price level is a significant determinant of money demand. As prices rise, the purchasing power of money decreases, leading individuals to demand more money to maintain their desired level of transactions. Conversely, when prices fall, the demand for money decreases as individuals require less cash for the same level of transactions.

Risk associated in alternative assets : If the alternative assets are risky, then the demand for money.

Technology of Payment and Receipt: Advancements in payment technologies, such as digital payment platforms like Paytm, can affect money demand. These technologies offer convenient and efficient alternatives to cash transactions, potentially reducing the need for holding physical currency. As electronic payment methods become more widespread and accessible, individuals may hold less cash, impacting money demand.

Theories of Money Demand

Quantity Theory of Money (Fisher Equation): This theory posits a direct relationship between the quantity of money in circulation and the price level in an economy. According to the Quantity Theory of Money, formulated by Irving Fisher, $MV = PT$, where M represents the money supply, V denotes the velocity of money (the rate at which money circulates), P stands for the price level, and T represents the volume of transactions. The theory suggests that changes in the money supply directly impact the price level, assuming a stable velocity of money and volume of transactions. The theory does not

specifically talk about the demand for money, but it helps us to find out the demand in equilibrium situation.

Cambridge Version of Quantity Theory of Money: The Cambridge version of the Quantity Theory of Money emphasizes the role of the demand for money in determining the price level. The theory establishes that the transaction demand for money is a fraction of nominal income. It can be used to find out the money supply in equilibrium situation. The theory is also known as neoclassical theory of demand for money.

Liquidity Preference Theory of Money (Keynesian Perspective): Proposed by John Maynard Keynes, the Liquidity Preference Theory of Money focuses on the demand for money as a store of wealth rather than solely a medium of exchange. According to this theory, individuals hold money not only for transactional purposes but also as a precautionary measure against uncertainty and for speculative purposes.

Baumol – Tobin Theory for Transactive Purpose: This theory helps to find out the optimum level of money to minimise the total cost made of transaction cost and opportunity cost.

Tobin's Theory for Speculative Money demand: This theory establishes a negative relationship between the interest rate and demand for money (i.e., cash) and positive relationship between the interest rate and the demand for bonds. According to Tobin, people hold cash and bonds both as oppose to Keynes whose view is that people either hold all cash or all bonds.

Friedman's Quantity Theory of Money : The theory sees money as an asset explains its demand in the same manner as the demand for any other good is explained.

5.6 KEY WORDS

- Bullish nature** : A bullish nature refers to a positive or optimistic outlook on the future performance of an asset or market. It suggests an expectation of rising prices and a favourable market environment.
- Bearish nature** : A bearish nature describes a negative or pessimistic outlook on the future performance of an asset or market. It implies an expectation of declining prices and an unfavourable market environment.
- Bond** : A bond is a debt investment where an investor loans money to an entity (typically a corporation or government) which borrows the funds for a defined period at a variable or fixed interest rate.
- Capital gain** : Capital gain refers to the profit earned from the sale of an asset, such as stocks, real estate, or bonds, which exceeds its purchase price.

Transactive motive for money demand : This motive refers to the need for money to facilitate day-to-day transactions such as buying goods and services. Individuals and businesses hold money to carry out their regular spending activities.

Precautionary motive for money demand : The precautionary motive for money demand refers to the desire to hold money as a buffer or reserve for unexpected future expenses or emergencies. It serves as a precaution against unforeseen circumstances.

Speculative motive for money demand : This motive involves holding money not for transactional purposes or precautionary reasons, but rather for potential investment opportunities. Individuals may hold money to take advantage of favorable market conditions or to quickly seize investment opportunities.

Velocity of money : Velocity of money is the rate at which money is exchanged in an economy. It is usually measured as the number of times money is spent in a given period. High velocity indicates a rapid circulation of money, while low velocity suggests slower economic activity.

5.7 SOME USEFUL REFERENCES

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Blanchard, O., & Johnson, D. R. (2012). Macroeconomics. Pearson.

Mankiw, N. G. (2009). Macroeconomics (7th ed.). Worth Publishers.

W, L. D. E. (1977). The demand for money: Theories and evidence (2nd ed.). Dun-Donnelley.

5.8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1. Money demand means the quantity of money (or cash) which households and firms want to hold with themselves, but not in other forms like bonds. Keeping other factors constant, just as the demand for a good is negatively related with its price, similarly, the demand for money is negatively related with interest rate.
2. Refer to Section 5.2

Check Your Progress 2

1. Refer the relationship between the quantity theory of money and demand for money in section 5.4.
2. Transactive motive, precautionary motive, speculative motive
3. Individuals make the portfolio of money as well as bond, and the speculative demand for money is a matter of uncertainty not the expectations about the interest rates as opposed to Keynes who used to think that individuals hold either all cash or all bonds depending upon their expectations of interest rate.
4. Refer to Section 5.4.4
5. According to Friedman, money demand can be studied in the same manner as the demand for any asset. The Friedman's 1970 version of the demand for money function: $M_d = f(Y_P, W, i_m, i_b, i_e, \pi_e, u)P$



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UNIT 6 MONEY SUPPLY

Structure

- 6.0 Objectives
- 6.1 Introduction
- 6.2 High-Powered Money and Money Supply
- 6.3 The Money-Multiplier Process
- 6.4 Factors Affecting the Money Multiplier and High-Powered Money
- 6.5 The Theory of Endogenous Money Supply
- 6.6 Let Us Sum Up
- 6.7 Key Words
- 6.8 Some Useful Books
- 6.9 Answers/Hints to Check Your Progress Exercises

6.0 OBJECTIVES

After going through the unit, you should be able to:

- Define high-powered money;
- Explain the link between high-powered money and the money supply;
- Explain the money-multiplier process;
- List the determinants of the money multiplier;
- Describe the factors which affect high-powered money; and
- Discuss the theory of endogenous money supply.

6.1 INTRODUCTION

In units 4 and 5, you learned about the nature and functions of money, and about the demand for money. In this unit, we will discuss supply of money. Till now, we have implicitly assumed that the supply of money is policy-determined. In an actual economy, however, the supply of money is determined jointly by the monetary authority and banks, and also in a way by the public. The central role in this is played by the monetary authority. We first need to be clear about the measure of money. Let us consider the narrow measure of money that we had mentioned in unit 4. That was M1 which was a sum of the currency and demand deposits. In this unit, we shall make use of another measure of money, but money that is created by the monetary authorities (Central Bank and the Government). This is called high-powered money.

In this unit, the discussion centres on how high-powered money, through a ‘money-multiplier’, determines the money supply. The unit discusses the nature of the high-powered money and its components. A discussion is provided about the nature of multiplier, its components and the relation among the components of the multiplier. You will come to understand why the multiplier is called ‘multiplier’. The unit discusses the factors which determine the money multiplier and high powered money. Finally, the unit discusses some approaches and views that suggest that money supply is not exogenous but is itself endogenously determined.

6.2 HIGH POWERED MONEY AND MONEY SUPPLY

In the realm of macroeconomics and monetary policy, the concepts of high-powered money and money supply play pivotal roles. High-powered money, also known as the monetary base or monetary base supply, represents the total amount of a country’s currency in circulation along with its reserves held by the central bank. Understanding high-powered money is crucial for comprehending the mechanisms through which central banks influence the broader money supply in an economy. This section aims to elucidate the concept of high-powered money, its components, and its implications on the broader money supply.

Let us study relation between high powered money and money supply with the help of the following equations:

$$M = C + D \quad (1)$$

where M= Money supply; C = Currency; D= Demand Deposit

$$H = C + R \quad (2)$$

$$\frac{H}{MB} = C + [RR + ER] \quad (3)$$

$$c^d = \frac{C}{D} \text{ or } C = c^d * D \quad (4)$$

$$rr = \frac{RR}{D} \text{ or } RR = rr * D \quad (5)$$

$$er = \frac{ER}{D} \text{ or } ER = er * D \quad (6)$$

where H = High Powered Money; C= Currency; R= Reserves; RR = Required Reserve; ER=Excess Reserve; MB= Monetary Base; c^d = Currency to Deposit ratio ; rr = required reserve to deposit ratio ; er = excess reserves to deposit ratio ; m= money multiplier

$$c^d = \frac{C}{D} \text{ or } C = c^d * D$$

$$rr = \frac{RR}{D} \text{ or } RR = rr * D$$

$$er = \frac{ER}{D} \text{ or } ER = er * D$$

$$M = m * H \text{ or } m = \frac{M}{H} \quad (7)$$

$$m = \frac{C + D}{C + [RR + ER]}$$

Dividing the above equation by D we get:

$$m = \frac{\frac{C}{D} + \frac{D}{D}}{\frac{C}{D} + \frac{RR}{D} + \frac{ER}{D}} \quad (8)$$

$$m = \frac{c^d + 1}{c^d + rr + er} \quad (9)$$

$$\text{since } m = \frac{M}{H}$$

$$\frac{M}{H} = \frac{c^d + 1}{c^d + rr + er} \quad (10)$$

$$M/H = c^d + 1 / c^d + rr + er \quad (11)$$

If there is no c^d ;

$$m = \frac{1}{rr + er}$$

If there is no c^d and er ;

$$m = \frac{1}{rr}$$

Components of High-Powered Money:

High-powered money comprises two main components:

- 1) **Currency in Circulation:** This component encompasses all the physical currency (notes and coins) circulating within an economy. It includes the money held by individuals, businesses, and institutions such as banks.
- 2) **Reserves Held by the Central Bank:** Reserves refer to the deposits that commercial banks hold at the central bank. These reserves are mandatory and serve as a buffer against financial crises and liquidity shortages. There are two types of reserves:

- *Required Reserves:* Mandated by central banks, these reserves are a fraction of a bank's deposits that it must hold in reserve either as cash in its vaults or as deposits with the central bank.
- *Excess Reserves:* These are reserves held by banks over and above the required amount. Banks may hold excess reserves for various reasons, including precautionary measures, meeting withdrawal demands, or regulatory compliance.

Implications of High-Powered Money on Money Supply

Understanding the relationship between high-powered money and the broader money supply is essential for comprehending monetary policy and its effects on the economy. The money supply refers to the total amount of money circulating in the economy, which includes both physical currency and various forms of bank deposits.

The process through which high-powered money influences the broader money supply can be elucidated through the money multiplier mechanism, which operates as follows:

- 1) **Money Multiplier Mechanism:** The central bank controls the monetary base through its open market operations, reserve requirements, and discount rates. By altering the monetary base, the central bank influences the overall money supply through a process called the money multiplier effect. The money multiplier represents the ratio of the change in the money supply to the change in the monetary base.
- 2) **Reserve Requirements:** When the central bank alters reserve requirements, it directly affects the amount of excess reserves that banks hold. Lowering reserve requirements increases excess reserves, providing banks with more capacity to extend loans, thereby expanding the money supply. Conversely, raising reserve requirements reduces excess reserves, limiting banks' lending capacity and contracting the money supply.
- 3) **Open Market Operations:** Through open market operations, the central bank buys or sells government securities in the open market. When the central bank purchases government securities, it injects funds into the banking system, increasing reserves and enabling banks to lend more, thus expanding the money supply. Conversely, selling government securities reduces reserves, constraining banks' ability to lend and contracting the money supply.
- 4) **Discount Rate:** The discount rate is the interest rate at which commercial banks can borrow from the central bank. When the central bank lowers the discount rate, borrowing becomes cheaper for commercial banks, encouraging them to borrow more, increase lending, and expand the money supply. Conversely, raising the discount rate

makes borrowing more expensive, leading to a decrease in lending and a contraction of the money supply.

High-powered money, comprising currency in circulation and reserves held by the central bank, serves as the foundation for the broader money supply in an economy. Understanding the relationship between high-powered money and the money supply is crucial for policymakers, economists, and students alike. By manipulating the components of high-powered money through various monetary policy tools, central banks can influence the money supply, thereby affecting economic activity, inflation, and interest rates. Thus, a thorough comprehension of high-powered money is essential for analyzing and formulating effective monetary policies.

Check Your Progress 1

- 1) What are the two main components of high-powered money?

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- 2) How does the central bank influence the money supply through open market operations?

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6.3 THE MONEY MULTIPLIER PROCESS

The money multiplier process is a fundamental concept in monetary economics, elucidating the mechanisms through which changes in the monetary base, controlled by central banks, influence the broader money supply in an economy. This section aims to provide a comprehensive understanding of the money multiplier process, its determinants, and its implications for monetary policy and economic stability.

Complete Deposit Multiplier

From equation (3), we know $\frac{H}{MB} = C + [RR + ER]$

$$\text{or } MB = c^d * D + rr * D + er * D$$

$$\text{or } MB = D[c^d + rr + er]$$

$$\text{or } D = \frac{MB}{c^d + rr + er}$$

Here $\frac{1}{c^d + rr + er}$ is complete deposit multiplier

If $c^d = 0$ and $er = 0$ then $\frac{1}{rr}$ is simple deposit multiplier

where MB= Monetary Base; c^d = Currency to Deposit ratio; rr = required reserve to deposit ratio; er = excess reserves to deposit ratio; D = Demand deposit

Complete Currency multiplier

$$c^d = \frac{C}{D}$$

$$C = c^d * D$$

We know from above that:

$$D = \frac{MB}{c^d + rr + er}$$

$$C = c^d * \frac{MB}{c^d + rr + er}$$

$$= \left[\frac{c^d}{c^d + rr + er} \right] * MB$$

Here $\frac{c^d}{c^d + rr + er}$ is a complete currency multiplier

Where MB= Monetary Base; c^d = Currency to Deposit ratio; rr = required reserve to deposit ratio; er = excess reserves to deposit ratio; D = Demand deposit

The Concept of the Money Multiplier

The money multiplier represents the ratio of the change in the money supply to the change in the monetary base. It illustrates how an initial injection of high-powered money (base money) into the banking system leads to a multiple expansion of the broader money supply through the process of bank lending and deposit creation.

Mechanisms of the Money Multiplier Process

The money multiplier process operates through a series of steps, as outlined below:

- 1) **Initial Injection of Base Money:** The process begins with the central bank injecting base money into the banking system through open market operations, discount window lending, or other monetary policy tools.
- 2) **Reserve Requirement:** Commercial banks are required to hold a certain percentage of their deposits as reserves, either as vault cash or as deposits with the central bank. This reserve requirement acts as a constraint on the amount of money banks can create through lending.